

VINEET PUNYAMOORTY

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EDUCATION

Purdue University

PhD in Electrical and Computer Engineering

West Lafayette, IN

Aug 2022 – Dec 2026 (expected)

Research Areas: **Machine Learning, Computer Vision, Multimodal Learning, Generative Models**

Selected Courses: Generative Models for Robotics, Diffusion Models, Artificial Intelligence

Indian Institute of Technology (IIT) Bombay

Bachelor of Technology & Master of Technology

Major: Electrical Engineering

Mumbai, India

Aug 2015 – May 2020

SELECTED PUBLICATIONS

- [1] Dynamic Obstacle Avoidance through Uncertainty-Based Adaptive Planning with Diffusion [[ArXiv](#)]
V. Punyamoorthy, et al. **IROS 2025**.
- [2] Contrastive Cross-Modal Learning on ECG and Chest X-Rays for Accessible Disease Prediction [[ArXiv](#)]
V. Punyamoorthy, et al. *Under Review* at **IEEE Transactions on Biomedical Engineering**.
- [3] Augmenting Generative Models with Biomedical Knowledge Graphs Improves Targeted Drug Discovery [[ArXiv](#)]
A. Malusare, **V. Punyamoorthy**, V. Aggarwal. *Accepted* to **IEEE Transactions on Artificial Intelligence**.
- [4] CarE-X: ECG-based Prediction of Cardiomegaly Metrics through Alignment with X-rays
V. Punyamoorthy, et al. *Under Review* at **IEEE Journal of Biomedical and Health Informatics**.
- [5] Privacy-aware Image Editing and Enhancement Pipeline
D. Tamboli, **V. Punyamoorthy**, V. Aggarwal. **U.S. Provisional Patent Application** No. 63/799,484 (2025)

WORK EXPERIENCE

Analyst

JP Morgan Chase & Co.

Oct 2020 – Jul 2022

Mumbai, India

- Designed and automated **ETL** data pipelines for large-scale portfolio datasets, integrating internal and external feeds from **8** vendors to produce a consolidated dataset for downstream analysis.
- Developed an interactive dashboard for performance analysis on large high-dimensional financial datasets, ensuring statistical consistency across $\sim$ **100K** records and delivering a **\$1.6M** annual impact.
- Delivered a real-time data pipeline by integrating multiple PostgreSQL databases, streaming data pipelines, and REST APIs, applying statistical methods to detect anomalies across **1M+** time-series records.
- Technical Skills: **Python, JavaScript, Pandas, SQL, Git, REST API, AWS, Agile, JIRA**

AWARDS AND HONORS

- **Undergraduate Research Award** in recognition of quality and extent of research in the Master's thesis (**2020**)
- Scholarship from the **Dutch Research Council (NWO)** for pursuing research at TU Delft, The Netherlands (**2018**)

PROJECTS

Visual Question Answering for Time-Series Data

Mar 2025 – Present

- Developed a **Vision-Language Model (VLM)**-based method for question answering on time-series data by aligning discrete token representations of signals with natural language using cross-modal alignment.

Enhancing Conditional Diffusion of Molecules using Knowledge Graphs

Dec 2024 – Present

- Designed a novel guidance mechanism for **Diffusion Models** to generate targeted drugs using conditioning information from biomedical knowledge graphs, improving protein-drug docking score by $\sim$ **10%**.

Uncertainty-based Diffusion Planning for Collision Avoidance

May 2024 – Sep 2024

- Developed a novel **diffusion model**-based method for adaptive re-planning in **robotic navigation** tasks using uncertainty estimates from a deep ensemble to reduce collisions, increasing mean trajectory length by **12.7%**.

SKILLS

Languages & Libraries	Python	C/C++	SQL	Bash Shell	PyTorch	TensorFlow	JAX
MLOps & Tools	AWS SageMaker	MLFlow	Docker	LangChain	ROS	MuJoCo	Git